

ASHWIN PRAKASH

☎ (781)-759-2993 ✉ ashwinprakash10@gmail.com 🌐 github.com/AshwinP10 📦 ashw.vercel.app/

Education

University of Texas at Austin

Major: BS Electrical and Computer Engineering, Minor: Robotics

May 2027

Austin, TX

Relevant Coursework: Algorithms, Software Testing, Software Lab, Data Science Lab, Software Design and Implementation, Probability and Random Processes, Embedded Systems, Linear Algebra, Discrete Mathematics – GPA: 3.7

Work Experience

Amazon

June 2026 – Present

SDE Intern - Computer Vision (Amazon Key, Devices)

Austin, Texas

- Building **computer-vision** pipelines to automate the installation-audit workflow for Amazon Key smart-access devices, parsing complex electrical schematics and field imagery to replace manual review for the Amazon Devices group.
- Training and fine-tuning custom detection and classification models (**YOLO**, **ResNet**) in **PyTorch** to verify correct device installation and flag wiring and component anomalies from real-world captures.
- Applying transfer learning, targeted data augmentation, and rigorous evaluation to harden **perception** accuracy under noisy conditions.

Autonomous Mobile Robotics Laboratory

September 2025 – Present

Robotics Software Engineer - Sensor Fusion & Perception

Austin, TX

- Implemented **sensor fusion** pipelines combining LiDAR, IMU, and camera data to enhance localization accuracy by **27%**.
- Implemented open-source **visual SLAM** and object tracking modules in ROS 2, reducing pose estimation drift by **18%**.
- Deployed real-time perception and control modules on **NVIDIA Jetson** for onboard autonomous navigation and obstacle avoidance.

Dell Technologies

June 2025 – August 2025

Software Engineering Intern

Round Rock, Texas

- Developed an AI-powered sprint reporting system using **LangChain** and Gemma, automating Jira analytics and reducing report generation time by **50%**.
- Built and deployed a **Random Forest Classifier** achieving **91%** accuracy for issue spillover prediction, improving triage efficiency.

Paradigm Robotics

January 2024 – May 2025

Software Engineer

Austin, TX

- Fine-tuned **YOLOv5** on a custom thermal+optical dataset using PyTorch, enabling reliable human detection in low-visibility environments for field robotics.
- Created real-time robot control GUI using **OpenCV**, WebSockets, and TCP/IP, deployed on embedded Linux Raspberry Pi.

Research Projects

VQA-Disagree: Multi-Model Disagreement for VQA Benchmark Stratification

[Code] | [Dataset]

- Built a multi-model disagreement pipeline in PyTorch using four 7B **vision-language models** (Qwen2-VL, LLaVA-1.6, InternVL2, MiniCPM-V), orchestrating GPU inference on Modal to auto-stratify 1,500 VQA items into difficulty buckets with zero human annotation.
- Validated the buckets with a Qwen2-VL-72B oracle (**4-bit QLoRA** on A100), confirming a **28.3 pp** Easy–Hard accuracy gap.
- Released a 456-sample difficulty-stratified VQA benchmark on HuggingFace; accepted as a workshop paper at **DataMFM @ CVPR 2026**.

DAVATaR - RobIN Lab Student Research

- Co-authored DAVATaR (submitted to **CoRL '26**), training multimodal **transformer policies** via **imitation learning** that fuse RGB vision, tactile, and immersion-depth signals for contact-rich manipulation across the air–water interface.
- Designed physically-grounded **visual augmentations** (reflection/refraction) that improved policy robustness to visual distribution shift, raising mean real-world task success from **28%** (vision-only) to **71%**.
- Built **behavior-cloning** training/eval pipelines in PyTorch (ResNet-18 spatial-softmax encoder + transformer fusion) and ran **closed-loop real-world rollouts** to benchmark policies and analyze failure modes.
- Teleoperated the robot to collect high-quality **multimodal demonstrations** (vision, tactile, depth), curating datasets with PhD mentors to improve policy sample efficiency.

Technical Skills

Languages: Python, C, C++, Java, SQL, JavaScript, HTML/CSS, VHDL, ARM

Libraries/Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLO, NumPy, pandas, ROS, Three.js, LangChain

Backend/Cloud/Tools: Node.js, Express, React, Django, Flask, FastAPI, Supabase, PostgreSQL, Docker, Kubernetes, AWS, Git, GitHub, Postman

Robotics/Perception: Sensor Fusion, Computer Vision, Imitation Learning, Behavior Cloning, Multimodal Transformers, Real-Time Systems, Embedded Linux

Operating Systems: Linux, Windows